

## BIOL 4559 Homework 04

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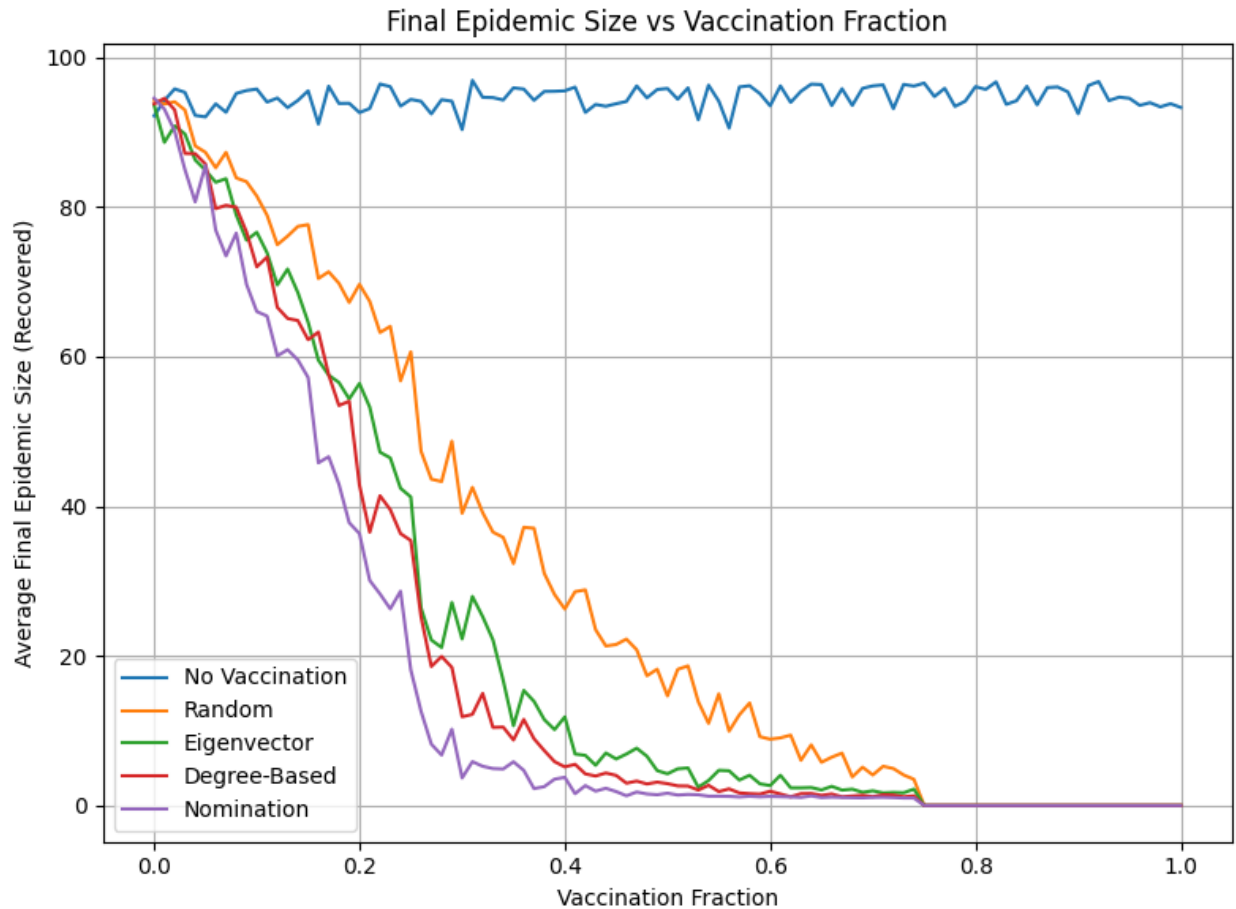
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### Question 1.

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In this problem, I want you to imagine that you are a policymaker with the future of the world (a generative network) in your hands. An epidemic has broken out and you need to handle it! You will be testing different vaccination strategies and comparing them against each other to see which is the best. The simulation should go as follows: (rest of the instruction are on Canvas).

**Solution Attempt** All of this is done in the included Jupyter Notebook. Here is the figure output:



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**Question 2.**

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After running your simulation, you will have to plot the number of recovered people for each strategy (look at sample notebook to understand what I mean) vs the proportion of vaccinated population.

- (a) Which vaccination strategy was most effective?

**Solution Attempt** From the figure, the most effective vaccination strategy is the degree-based strategy, closely followed by the eigenvector-based strategy. These two methods consistently achieve the smallest final epidemic size for any given vaccination fraction and drive the epidemic to near-zero spread at substantially lower vaccination levels than the other strategies. In contrast, random vaccination performs significantly worse, while nomination vaccination shows intermediate performance.

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- (b) If you were the policymaker, which strategy would and why? Consider cost, data availability, and feasibility?

**Solution Attempt** As a policymaker, I would adopt the degree-based vaccination strategy. While eigenvector-based vaccination performs similarly, degree-based vaccination is far more practical because node degrees can be estimated using simple contact tracing and interaction counts, whereas eigenvector centrality requires global knowledge of the network structure and expensive computation. Degree-based vaccination therefore offers an excellent balance between epidemic control effectiveness, data availability, and real-world feasibility.

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- (c) If one vaccination strategy outperformed the others, explain why. What features of the network structure or selection method made that strategy more effective?

**Solution Attempt** The superior performance of the degree-based and eigenvector-based strategies is a direct consequence of the heavy-tailed structure of the Barabási–Albert network. In scale-free networks, a small number of high-degree hub nodes are responsible for a disproportionately large fraction of transmission pathways. Vaccinating these hubs early dramatically fragments the network and prevents large-scale cascades. Random vaccination, by contrast, wastes vaccines on low-degree peripheral nodes that contribute little to disease spread, which explains its poor performance. Nomination vaccination partially targets hubs but does so imperfectly, leading to intermediate effectiveness.

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- (d) How do you expect each vaccination strategy to perform if we simulated the epidemic on an Erdos-Renyi network? Justify your logic and include the same figure as before for this network.

**Solution Attempt** On an Erdős–Rényi network, all vaccination strategies would perform much more similarly because the degree distribution is narrow and lacks dominant hub nodes. Since most nodes have approximately the same number of connections, targeted strategies such as degree-based and eigenvector-based vaccination lose their advantage. As a result, random vaccination would become nearly as effective as targeted strategies,

and the dramatic separation between curves observed in the Barabási–Albert case would largely disappear.

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